

Question 1
Not yet answered
Marked out of 8.00
Flag question

The following constrained minimisation problem is considered:

minimize $f(\mathbf{x}) = (-x_1 + 1)^2 + 3x_2 + 2x_1^2$

Subject to the inequality constraints:

$g_1(\mathbf{x}) = -x_1 + x_2 + 4 \leq 0$

$g_2(\mathbf{x}) = 3 - x_1 \leq 0$

Determine the optimum solution to the problem using the KKT necessary conditions and/or graphical methods.

Select one:

- ☐ a. $x_1 = 3, x_2 = -2.5$
- ☒ b. $x_1 = 3, x_2 = -1$
- ☐ c. $x_1 = 1, x_2 = -2$
- ☐ d. $x_1 = -1, x_2 = -2$
- ☐ e. None of the other solutions
- ☒ f. $x_1 = 3, x_2 = -2$
- ☐ g. $x_1 = 3, x_2 = -3$
- ☐ h. $x_1 = 1.5, x_2 = -2$

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$\{y_1 = -2, y_2 = 6, x_1 = 3, x_2 = -1\}$

Question 2
Not yet answered
Marked out of 3.00
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We will consider gradient-based minimisation of the following unconstrained function:

$f(\mathbf{x}) = -2x_1 + 3x_2 + (x_1 - 2)^2 + (1 - x_2)^2$

The starting point is: $\mathbf{x}^{(0)} = [1, 2]^T$.

In order to complete the first iteration of the steepest descent method for the function using the analytical approach, the gradient must be evaluated. What is the expression for the gradient at the starting point, $\mathbf{c}^{(0)}$?

Select one:

- ☒ a. $\mathbf{c}^{(0)} = [2x_1 - 6, 2x_2 + 1]^T$
- ☐ b. $\mathbf{c}^{(0)} = [2x_1 - 6, 1 - 2x_2]^T$
- ☐ c. None of the other solutions
- ☐ d. $\mathbf{c}^{(0)} = [2x_1 + 1, 2x_2 + 2]^T$
- ☐ e. $\mathbf{c}^{(0)} = [2x_1 + 2, 2x_2 + 1]^T$
- ☐ f. $\mathbf{c}^{(0)} = [2x_1 - 6, x_2 + 2]^T$
- ☐ g. $\mathbf{c}^{(0)} = [2x_1 + 6, 1 + 2x_2]^T$
- ☐ h. $\mathbf{c}^{(0)} = [2x_1 - 6, x_2 - 3]^T$

$$\mathbf{c}_0 = \begin{bmatrix} 2x_1 \\ 2x_2 \end{bmatrix}$$

$$\mathbf{c}_0 = \begin{bmatrix} -6 + 2x_1 \\ 1 + 2x_2 \end{bmatrix}$$

Question 3
Not yet answered
Marked out of 5.00
Flag question

In order to complete the first iteration of the steepest descent method for the function above using the analytical approach, the 1D line search problem should be established.

What is the expression for $f(\alpha)$, where α is the step size in direction $\mathbf{d}^{(0)}$?

Select one:

- ☐ a. $41\alpha^2 + 41\alpha + 6$
- ☐ b. $45\alpha^2 - 45\alpha + 15$
- ☐ c. $3\alpha^2 - 5\alpha - 3$
- ☐ d. None of the other solutions
- ☐ e. $13\alpha^2 + 13\alpha - 1$
- ☐ f. $3\alpha^2/4 - 3\alpha/2 - 3$
- ☒ g. $41\alpha^2 - 41\alpha + 6$
- ☐ h. $13\alpha^2 - 13\alpha - 1$

$$f_0 := \text{eval}\left(f(x), \left[x_1 = \alpha(1), x_2 = \alpha(2)\right]\right)$$

$$\text{expand}$$

$$f_0 := -23\alpha + 4 + (4\alpha - 1)^2 + (-1 + 5\alpha)^2$$

$$41\alpha^2 - 41\alpha + 6$$

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Question 4
Not yet answered
Marked out of 8.00
Flag question

We will consider gradient-based minimisation of the following unconstrained function:

$f(\mathbf{x}) = 3x_1^2 - 2x_2 + (1 - x_1)^2 + (2 - x_2)^2$

The starting point is: $\mathbf{x}^{(0)} = [1, 2]^T$.

You should complete the first iteration of the modified Newton's method.

What are the design variables after the first iteration, $\mathbf{x}^{(1)}$?

Select one:

- ☐ a. $\mathbf{x}^{(1)} = [7/4, 1]^T$
- ☐ b. $\mathbf{x}^{(1)} = [1/2, -3]^T$
- ☐ c. $\mathbf{x}^{(1)} = [-7/37, -84/37]^T$
- ☐ d. None of the other solutions
- ☐ e. $\mathbf{x}^{(1)} = [1/2, 3]^T$
- ☐ f. $\mathbf{x}^{(1)} = [7/37, 84/37]^T$
- ☐ g. $\mathbf{x}^{(1)} = [-1/4, 3]^T$
- ☒ h. $\mathbf{x}^{(1)} = [1/4, 3]^T$

Previous page

$$f(\mathbf{x}) = 0.9, 4.3$$

Next page

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Step 6: Operator verifier

$$x_1 = x_0 + \alpha_0 \cdot d_0$$

$$x_1 = \begin{bmatrix} 1 \\ 4 \\ 3 \end{bmatrix}$$

Question 5
Not yet answered
Marked out of 2.00
Flag question

An non-linear optimisation problem is given by:

minimise $f(\mathbf{x}) = 4x_1^2 + 3x_2^2 - 5x_1x_2 - 8$

Subject to the constraints:

$g_1(\mathbf{x}) = -x_1 + x_2 - 4 \leq 0$

$g_2(\mathbf{x}) = \frac{1}{x_1} - x_2 \leq 0$

Determine the gradient of the objective function and constraints at the point $\mathbf{x}_0 = [2, 2]^T$.

Select one:

- ☐ a. None of the other solutions
- ☒ b. $\nabla f(\mathbf{x}_0) = \begin{bmatrix} 6 \\ 2 \end{bmatrix}$, $\nabla g_1(\mathbf{x}_0) = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$, $\nabla g_2(\mathbf{x}_0) = \begin{bmatrix} -1/4 \\ -1 \end{bmatrix}$
- ☐ c. $\nabla f(\mathbf{x}_0) = \begin{bmatrix} 8 \\ -2 \end{bmatrix}$, $\nabla g_1(\mathbf{x}_0) = \begin{bmatrix} 1/2 \\ -2 \end{bmatrix}$, $\nabla g_2(\mathbf{x}_0) = \begin{bmatrix} 1/2 \\ 2 \end{bmatrix}$
- ☐ d. $\nabla f(\mathbf{x}_0) = \begin{bmatrix} 8 \\ -2 \end{bmatrix}$, $\nabla g_1(\mathbf{x}_0) = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$, $\nabla g_2(\mathbf{x}_0) = \begin{bmatrix} -1/4 \\ -1 \end{bmatrix}$
- ☐ e. $\nabla f(\mathbf{x}_0) = \begin{bmatrix} 6 \\ 2 \end{bmatrix}$, $\nabla g_1(\mathbf{x}_0) = \begin{bmatrix} -2 \\ 2 \end{bmatrix}$, $\nabla g_2(\mathbf{x}_0) = \begin{bmatrix} 1/2 \\ -2 \end{bmatrix}$
- ☐ f. $\nabla f(\mathbf{x}_0) = \begin{bmatrix} 8 \\ -2 \end{bmatrix}$, $\nabla g_1(\mathbf{x}_0) = \begin{bmatrix} -2 \\ 2 \end{bmatrix}$, $\nabla g_2(\mathbf{x}_0) = \begin{bmatrix} 1/2 \\ -2 \end{bmatrix}$

$$\nabla f(\mathbf{x}) = \begin{bmatrix} 8x_1 - 5x_2 \\ 6x_2 - 5x_1 \end{bmatrix}$$

$$\nabla g_1(\mathbf{x}) = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$\nabla g_2(\mathbf{x}) = \begin{bmatrix} -1/x_1^2 \\ -1 \end{bmatrix}$$

$$\nabla f(\mathbf{x}_0) = \begin{bmatrix} 6 \\ 2 \end{bmatrix}$$

$$\nabla g_1(\mathbf{x}_0) = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$\nabla g_2(\mathbf{x}_0) = \begin{bmatrix} -1/4 \\ -1 \end{bmatrix}$$

$$\nabla f(\mathbf{x}) = \begin{bmatrix} 8x_1 - 5x_2 \\ 6x_2 - 5x_1 \end{bmatrix}$$

$$\nabla g_1(\mathbf{x}) = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$\nabla g_2(\mathbf{x}) = \begin{bmatrix} -1/x_1^2 \\ -1 \end{bmatrix}$$

$$\nabla f(\mathbf{x}_0) = \begin{bmatrix} 6 \\ 2 \end{bmatrix}$$

$$\nabla g_1(\mathbf{x}_0) = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$\nabla g_2(\mathbf{x}_0) = \begin{bmatrix} -1/4 \\ -1 \end{bmatrix}$$

Question 6

Answer saved
Marked out of 1.00
Flag question

The above problem is to be solved by a SiP approach. Determine the linearised subproblem when $\mathbf{x}_0 = [2 \ 2]^T$

Example 12.3

Select one:

- ☐ a. minimise $\bar{f} = 6d_1 + 2d_2$, subject to: $\begin{bmatrix} -1 & -1/4 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} d_1 \\ d_2 \end{bmatrix} \leq \begin{bmatrix} -4 \\ -3 \end{bmatrix}$
- ☐ b. minimise $\bar{f} = 6d_1 + 2d_2$, subject to: $\begin{bmatrix} -1 & 1 \\ -1/4 & -1 \end{bmatrix} \begin{bmatrix} d_1 \\ d_2 \end{bmatrix} \leq \begin{bmatrix} -4 \\ -3 \end{bmatrix}$
- ☒ c. minimise $\bar{f} = 8d_1 - 2d_2$, subject to: $\begin{bmatrix} -1 & -1/4 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} d_1 \\ d_2 \end{bmatrix} \leq \begin{bmatrix} 4 \\ -3 \end{bmatrix}$
- ☐ d. minimise $\bar{f} = 6d_1 + 2d_2$, subject to: $\begin{bmatrix} -2 & 1/2 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} d_1 \\ d_2 \end{bmatrix} \leq \begin{bmatrix} -4 \\ -3 \end{bmatrix}$
- ☐ e. minimise $\bar{f} = 6d_1 + 2d_2$, subject to: $\begin{bmatrix} -1 & 1 \\ -1/4 & -1 \end{bmatrix} \begin{bmatrix} d_1 \\ d_2 \end{bmatrix} \leq \begin{bmatrix} 4 \\ 3 \end{bmatrix}$
- ☐ f. minimise $\bar{f} = 8d_1 - 2d_2$, subject to: $\begin{bmatrix} -1 & 1 \\ -1/4 & -1 \end{bmatrix} \begin{bmatrix} d_1 \\ d_2 \end{bmatrix} \leq \begin{bmatrix} -4 \\ -3 \end{bmatrix}$
- ☐ g. None of the other solutions
- ☐ h. minimise $\bar{f} = 6d_1 + 2d_2$, subject to: $\begin{bmatrix} -2 & 1/2 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} d_1 \\ d_2 \end{bmatrix} \leq \begin{bmatrix} 4 \\ 3 \end{bmatrix}$

Lesson 8

$$\underline{A} = \begin{bmatrix} \nabla g_1 & \nabla g_2 \end{bmatrix}$$

$$\underline{b}^T = \begin{bmatrix} -g_1 & -g_2 \end{bmatrix}$$

$$x_2 = 2$$

$$x_2 = 2$$

$$x_2 = 2$$

$$x_2 = 2$$

$$d = \begin{bmatrix} \theta^T x_1 & \theta^T x_2 \\ \theta^T x_2 & \theta^T x_2 \end{bmatrix}$$

$$d = \begin{bmatrix} -1 & -1/4 \\ 1 & -1 \end{bmatrix}$$

$$b = \begin{bmatrix} -g_1 \\ -g_2 \end{bmatrix}$$

$$b = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

$$A \begin{bmatrix} d_1 \\ d_2 \end{bmatrix} \leq b$$

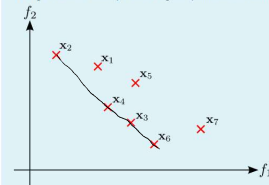
$$\begin{bmatrix} -1 & -1/4 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} d_1 \\ d_2 \end{bmatrix} \leq \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

f

Question 7

Not yet answered
Marked out of 4.00
Flag question

This is a question where multiple solutions may apply. Correct solutions yield positive points, while wrong solutions count negatively. The total score of this sub-question cannot be negative. In the figure below are shown points from a given optimisation. It is informed that the non-dominated points are part of the Pareto front.



Determine which of the points that are Pareto optimal:

Select one or more:

- ☐ a. x_1
- ☒ b. x_2
- ☒ c. x_3
- ☒ d. x_4
- ☐ e. x_5
- ☒ f. x_6
- ☐ g. x_7
- ☐ h. None of points are Pareto optimal

Designspace er fri. Feasible criterion er f_1 , da ingen constraints skal x_2 og x_6 med?

Jeg vælger at tage x_2, x_4, x_3 og x_6 med, da det alle er gode kompromis, hvor i hvert fald en funktion følger og den anden stiger

Question 8

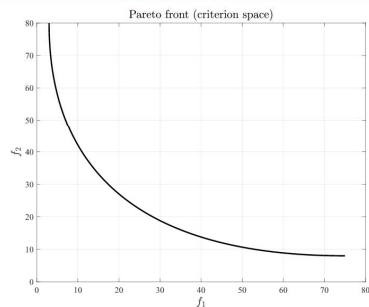
Not yet answered
Marked out of 2.00
Flag question

A multi-objective optimisation problem is given by:

$$\text{minimise } f_1(x_1, x_2) = (x_1 - 8)^2 + (x_2 - 1)^2 + 3$$

$$f_2(x_1, x_2) = (x_1 - 2)^2 + (x_2 - 7)^2 + 8$$

The figure below shows the Pareto front for the problem.



Determine the utopia point.

Select one:

- ☐ a. (8, 1) and (2, 7)
- ☐ b. None of the other solutions
- ☐ c. (8, 3)
- ☐ d. (80, 75)
- ☐ e. (75, 80)
- ☒ f. (3, 8)
- ☐ g. (0, 0)

$$\begin{bmatrix} 3 \\ 8 \end{bmatrix}$$

$$f_{1, \min} = f_1(8, 1) = 3$$

$$f_{2, \min} = f_2(2, 7) = 8$$

$$= \text{Utopia} = \begin{bmatrix} 3 \\ 8 \end{bmatrix}$$

Question 9

Not yet answered
Marked out of 2.00
Flag question

Assume that the multi-objective problem is solved as a single objective problem, $U(\mathbf{x})$, using the weighted sum method with $w_1 = 4$ and $w_2 = 3$. Determine the optimum set of design variables $\mathbf{x}^*$ and the minimum objective function value $U(\mathbf{x}^*)$.

Select one:

- ☐ a. $\mathbf{x}^* = [5 \ 4]^T \Rightarrow U(\mathbf{x}^*) = 47$
- ☐ b. $\mathbf{x}^* = [5 \ 4]^T \Rightarrow U(\mathbf{x}^*) = 11$
- ☐ c. $\mathbf{x}^* = \begin{bmatrix} 38 \\ 7 \end{bmatrix}^T \Rightarrow U(\mathbf{x}^*) = \frac{866}{7}$
- ☒ d. $\mathbf{x}^* = \begin{bmatrix} 38 \\ 7 \end{bmatrix}^T \Rightarrow U(\mathbf{x}^*) = \frac{1118}{7}$
- ☐ e. $\mathbf{x}^* = [5 \ 4]^T \Rightarrow U(\mathbf{x}^*) = 36$
- ☐ f. $\mathbf{x}^* = [0 \ 0]^T \Rightarrow U(\mathbf{x}^*) = 11$
- ☐ g. None of the other solutions

$$\alpha = \text{solve}(\text{eq}, [x_1, x_2])$$

$$\alpha = \begin{bmatrix} x_1 = \frac{38}{7}, x_2 = \frac{25}{7} \end{bmatrix}$$

$$x_1 = \text{eval}(\text{fval}(\alpha(1)))$$

$$x_1 = 5.428571429$$

$$x_2 = \text{eval}(\text{fval}(\alpha(2)))$$

$$x_2 = 3.571428571$$

$$\text{Minimum} = \text{eval}(U)$$

$$\text{Minimum} = 159.4285714$$

Question 10
Not yet answered
Marked out of 2.00
Flag question

This is a question where multiple solutions may apply. Correct answers yield positive points, while wrong solutions count negatively. The total score of this sub-question cannot be negative.
Determine, for the above problem where the multi-objective problem is solved as a single objective problem, which of the following statements that are correct:

Select one or more:

- ☒ a. The found optimum is a global optimum
☐ b. None of the other statements are correct
☒ c. The found optimum is a minimum
☐ d. The found optimum is a maximum
☐ e. The found optimum is the utopia point
☐ f. The found optimum is a saddle point

Da ingen constraints og 6 cirkulære funktioner
det er også et minimum?
I tvivl

Question 11
Not yet answered
Marked out of 2.00
Flag question

An optimisation problem is formulated as:

$$\text{minimise } f(\mathbf{x}) = 0.02x_1^2 - 0.2x_1x_2 + 0.08x_2^2$$

Subject to:

$$\begin{aligned} g_1(\mathbf{x}) &= -x_1 + 2x_2 \leq 0 \\ g_2(\mathbf{x}) &= -x_1 - x_2 + 10 \leq 0 \end{aligned}$$

The initial starting point for an optimisation is $\mathbf{x}^{(0)} = [8, 3]^T$, for which a search direction is determined to $\mathbf{d}^{(0)} = [1.4, -2.4]^T$ and the vector of Lagrange multipliers for the constraints is $\mathbf{u} = [0, 4.64]^T$.

To determine the step size the inexact line search procedure should used, where the initial settings are $R_0 = 1$ and $\gamma = 0.5$.
Choose the trial step according to the sequence $t_0 = 1, t_1 = \frac{1}{2}, t_2 = \frac{1}{4}, t_3 = \frac{1}{8}, \dots$

Determine the point for the first trial step (i.e. for $j = 0$).

Select one:

- ☐ a. None of the other solutions
☐ b. $\mathbf{x}^{(1,0)} = [8, 0.68]^T$
☐ c. $\mathbf{x}^{(1,0)} = [6.6, 5.4]^T$
☒ d. $\mathbf{x}^{(1,0)} = [9.4, 0.6]^T$
☐ e. $\mathbf{x}^{(1,0)} = [7.3, 4.2]^T$
☐ f. $\mathbf{x}^{(1,0)} = [8, 7.64]^T$
☐ g. $\mathbf{x}^{(1,0)} = [8.7, 1.8]^T$

Trialstep 1: $j=0, i,j=1,0=1$
Updated trialstep
 $x_{0,1} = x_0 + \gamma_0 d_0$

$$x_{0,1} = \begin{bmatrix} 9.40000000000000 \\ 0.600000000000000 \end{bmatrix}$$

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Question 12
Not yet answered
Marked out of 2.00
Flag question

What will be the value of Pshenichny's descent function for the first trial step?

Select one:

- ☐ a. $\Phi_{1,0} \approx 9.37$
☐ b. $\Phi_{1,0} \approx 24.79$
☐ c. $\Phi_{1,0} \approx 11.24$
☒ d. $\Phi_{1,0} \approx 15.49$
☐ e. None of the other solutions
☐ f. $\Phi_{1,0} \approx 12.31$

$$\Phi_{0,1} = f_0 + \gamma_0^T d_{0,1}$$

$$\Phi_{0,1} = 15.4940480000000$$

Question 13
Not yet answered
Marked out of 3.00
Flag question

For which trial step is the descent condition first fulfilled?

Select one:

- ☐ a. $t_0 = \frac{1}{4}$
☐ b. None of the other solutions
☐ c. The inexact line search does not converge
☐ d. $t_0 > t_1$
☐ e. $t_0 = 1$
☐ f. $t_2 = \frac{1}{4}$
☒ g. $t_1 = \frac{1}{2}$

Trialstep 2: $j=1, i,j=1,1=0.5$
 $x_{1,1} = x_0 + t_1 d_0$

$$t_2$$

$$x_{1,1} = \begin{bmatrix} 8.70000000000000 \\ 1.80000000000000 \end{bmatrix}$$

$$0.5$$

Question 14
Not yet answered
Marked out of 3.00
Flag question

What is the descent function value for the trial step that fulfils the descent condition?

Select one:

- ☐ a. None of the other solutions
☒ b. $\Phi_{1,1} \approx 10.88$
☐ c. $\Phi_{1,1} \approx 9.64$
☐ d. $\Phi_{1,1} \approx 8.31$
☐ e. $\Phi_{1,1} \approx 4.32$
☐ f. $\Phi_{1,1} \approx 10.29$

$\nabla f = 0$, ingen constraints violated
 $\Phi_{1,1} = f_1 + \gamma_1^T d_{1,1}$

$$\Phi_{1,1} = 10.8778880000000$$

Question 15
Not yet answered
Marked out of 10.00
Flag question

Select one or more of the following statements. Correct answers yield positive points, while wrong answers count negatively. Total score of this sub-question can not be negative.
We assume you are going to solve a design optimization problem where the maximum pressure of fluid running in a network of pipes is to be minimized. The pipe system consists of $n = 12$ pipes, the design variables of the problem are denoted $x_i, i = 1, \dots, n$, and these are the cross sectional areas of the pipes with given nonzero lower and upper bounds.
We assume that you have been given an analysis model that can compute the pressure distribution in the network of pipes, and this will give you pressure values $P_i, i = 1, \dots, n$, related to the pipes. Gradients can also be computed by the analysis code.
The following statements about the optimization problem of minimizing the maximum pressure in the network of pipes are true:

Select one or more:

- ☐ a. If the maximum pressure in the network of pipes is chosen as objective function in each iteration, then the best solution to the optimization problem will always be obtained, as no approximation to the objective function is introduced.
☐ b. The pressure computed in a given pipe is assumed to be a continuous and differentiable function of the design variables, but the min-max optimization problem is in general not continuous and differentiable. *Et er jo ikke et differentiable, i hvert et de andre pletter.*
☒ c. If the min-max problem is solved using a p-norm function approach, where the objective is computed using a p-norm function with a penalization power $p=5$ and all n pressure values are included, then the objective function underestimates the true objective function. *ledsæt 9 side se max norm*
☒ d. The min-max problem can be solved efficiently by the use of a bound formulation approach in a gradient based optimization algorithm.
☐ e. The pressure computed in a given pipe is assumed to be a continuous and differentiable function of the design variables, and thus the min-max optimization problem is also continuous and differentiable.
☒ f. The min-max problem can be solved efficiently by the use of aggregate functions in a gradient based optimization algorithm. *you say you use aggregation functions 3 min under i forelesning 9 part 4*

i tvivl

- Including very many stress values in one of these evaluating functions (also called aggregate functions) does not work as the importance of the highest stresses becomes too small. This is especially seen for the weighted sum method.
- The choice of penalization power p for the weighted global criterion method is important! In order to give preference to the highest values, a value of p being relatively high (for example 5.0) may be advantageous. Choosing a too high penalization power makes the optimization problems more nonlinear and difficult to solve.

Question 16
Not yet answered
Marked out of 5.00
Flag question

Select one or more of the following statements. Correct answers yield positive points, while wrong answers count negatively. Total score of this sub-question can not be negative.
Gradient based optimization requires the computation of derivatives (sensitivities), and in case of implicit criterion functions, as studied in the structural optimization part of the course, the computational cost depends on the chosen method for computing sensitivities. We consider a structural optimization problem of a steel component where the objective is to minimize the von Mises stress at a given point, subject to a mass constraint. A total of 20 design variables are defined. The problem will be solved using a gradient based approach.
The following statements about this problem are true:

Select one or more:

- ☒ a. The adjoint method is computationally more efficient than the direct differentiation method for obtaining sensitivities of the objective function.
☐ b. The direct differentiation method is computationally more efficient than the adjoint method for obtaining sensitivities of the objective function.
☐ c. The von Mises stress at a given location in a finite element model of a steel component is normally a continuous and differentiable function.
☐ d. The overall finite difference method will always be the computationally most efficient approach for obtaining sensitivities.

A gradient to funktionen, men mange designvariable

Question 17
Not yet answered
Marked out of 5.00
Flag question

An optimization problem is defined via the following objective function

$$f(\mathbf{x}) = \frac{1}{2} \sum_{i=1}^3 (x_1(1-x_2^2) - c_i)^2$$

where $\mathbf{c} = [1.5, 2.25, 2.625]^T$ and $\mathbf{x}_0 = [0.1, 0.1]^T$.

Select one or more of the following statements. Correct answers yield positive points, while wrong answers count negatively. Total score of this sub-question can not be negative.

Select one or more:

- ☒ a. The problem has two variables
☐ b. The problem has three variables
☐ c. This is a least-square problem, which may be solved using a Gauss-Newton formulation
☐ d. The residual vector contains two elements
☒ e. The residual vector contains three elements
☐ f. The residual vector contains four elements
☐ g. The residual vector contains only one element
☐ h. We are missing an equation, in order to solve the problem using a Gauss-Newton formulation.
☐ i. None of the other answers

$$x_0 = \begin{bmatrix} 0.1 \\ 0.1 \end{bmatrix}$$

$$r_i = f_3$$

husk dette!

$$r = x_1(1-x_2^2) - c_i \quad \nabla r = \begin{bmatrix} 1-x_2^2 \\ -2x_1x_2 \end{bmatrix}$$

Question 18
Not yet answered
Marked out of 4.00
Flag question

An optimization problem is defined via the following objective function

$$f(\mathbf{x}) = \frac{1}{2} \sum_{i=1}^3 (x_1(1-x_2^2) - c_i)^2$$

where $\mathbf{c} = [1.5, 2.25, 2.625]^T$ and $\mathbf{x}_0 = [0.1, 0.1]^T$.

Calculate the initial numerical value of the objective function i.e. $f(\mathbf{x}_0)$.

Select one:

$$f =$$

Discrete approach to Design Sensitivity Analysis (DSA)

The remaining part of DSA of state variables concentrates on the discrete approach.

The starting point is the governing equations in residual form, stated here again for convenience:

$$\mathbf{R}(\mathbf{U}(\mathbf{x}), \mathbf{x}) = \mathbf{0} \quad \text{if solved by Newton's method: } \left[\frac{\partial \mathbf{R}(\mathbf{U}^h)}{\partial \mathbf{U}} \right] \Delta \mathbf{U}^h = -\mathbf{R}(\mathbf{U}^h) \quad (25)$$

i.e. the derivative $\partial \mathbf{U} / \partial \mathbf{x}_i$ is defined implicitly through the residual $\mathbf{R}$ (we need to solve a coupled system of equations).

The discrete approach can be applied using either

- the direct differentiation method (DDM) many design variables dyrt
- the adjoint method (AM), seenet for 3 funktioner

Both these methods are described in the following.

Department of Mechanics and Production

NEW LINK: Optimization Course 2020
Lecture 8 - How to use adjoint analysis to design sensitivity analysis

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Comparing the Direct Differentiation Method and the Adjoint Method

To summarize,

- the direct differentiation method (DDM) involves the solution of Eq. (28) for each design variable x_i
- the adjoint method (AM) requires the solution of Eq. (31) for each general response function $\hat{f}$. *is funktioner for f, men mange designvariable*

The most efficient DSA method to use thus depends on the problem considered, i.e. depending on the number of design variables compared to the number of criterion functions.

The Jacobian (the system matrix) from the analysis can be reused which in most cases makes it possible to perform the sensitivity analysis much faster than the analysis.

For the adjoint method the Jacobian has to be symmetric in order to be reused.

Note: In case of using explicit solution schemes where the use of the Jacobian is avoided (like explicit time integration schemes), analysis and DSA for just one design variable are equally time and memory consuming.

The use of DSA will be illustrated by an example in the following.

Marked out of 4.00
Flag question

$f(x) = \frac{1}{2} \sum_{i=1}^3 (x_i(1-x_i) - c_i)^2$
where $c=[1.5, 2.25, 2.625]^T$ and $x_0=[0.1, 0.1]^T$.
Calculate the initial numerical value of the objective function i.e. $f(x_0)$.

- Select one:
- ☐ a. -12.991
- ☐ b. 1.41
- ☐ c. 2.991
- ☐ d. 0.705
- ☐ e. 12.991
- ☐ f. [-1.41, -2.151, -2.5251]^T
- ☒ g. 6.496
- ☐ h. None of the other solutions

$f =$
6.4965

time integration schemes), analysis and DSA for just one design variable are equally time and memory consuming.

The use of DSA will be illustrated by an example in the following.

Question 19
Not yet answered
Marked out of 5.00
Flag question

An optimization problem is defined via the following objective function

$$f(x) = \frac{1}{2} \sum_{i=1}^3 (x_i(1-x_i) - c_i)^2$$

where $c=[1.5, 2.25, 2.625]^T$ and $x_0=[0.1, 0.1]^T$.
The 3rd row of the Jacobian matrix is equal to:

- Select one:
- ☐ a. $(x_1(1-x_2)-c_1)(1-x_2) \quad (x_1(1-x_2)-c_2)(1-x_2) \quad (x_1(1-x_2)-c_3)(1-x_2)$
- ☐ b. $1-x_2 \quad -x_1$
- ☐ c. $c_1(1-x_2) \quad c_2(1-x_2) \quad c_3(1-x_2)$
- ☒ d. $1-x_2^2 \quad -3x_1x_2^2$
- ☐ e. $c_1(1-x) \quad c_2(-3x_1x_2^2)$
- ☐ f. None of the other solutions
- ☐ g. $1-x_2 \quad 1-x_2^2 \quad 1-x_2^2$
- ☐ h. $-x_1c_1 \quad -x_12x_2c_2 \quad -x_23x_1^2c_3$

$$i = 3 \quad \begin{bmatrix} 1-x_2^2 \\ -x_1x_2^2 \end{bmatrix} \Rightarrow \begin{bmatrix} 1-x_2^2 \\ -3x_1x_2^2 \end{bmatrix}$$

Question 20
Not yet answered
Marked out of 5.00
Flag question

An optimization problem is defined via the following objective function

$$f(x) = \frac{1}{2} \sum_{i=1}^3 (x_i(1-x_i) - c_i)^2$$

where $c=[1.5, 2.25, 2.625]^T$ and $x_0=[0.1, 0.1]^T$.
Completed the first 4 iterations using the Levenberg Marquardt algorithm for step size regulation. Using an initial lambda equal to 0.02 ($\lambda=0.02$).
Which of the following 8 answers represent the state of the solver at the 4th iteration?

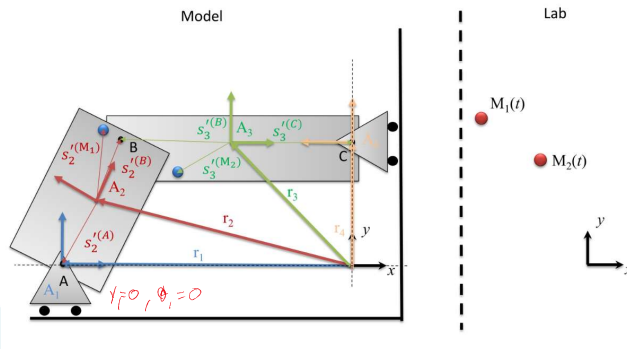
- Select one:
- ☐ a. $x_k=[-2.3277, 1.2434]^T \quad f(x_k)=1.0292$ and $\lambda=0.02$
- ☐ b. Failed to solve the problem
- ☐ c. $x_k=[3.0, 0.5]^T \quad f(x_k)=2.2857e-14$ and $\lambda=0.00125$
- ☐ d. None of the other solutions
- ☐ e. $x_k=[-2.1837, 1.2948]^T \quad f(x_k)=0.7446$ and $\lambda=0.00125$
- ☒ f. $x_k=[2.9847, 0.4966]^T \quad f(x_k)=2.1415e-05$ and $\lambda=0.01$
- ☐ g. $x_k=[3.0, 0.5]^T \quad f(x_k)=0.0$ and $\lambda=0.00125$
- ☐ h. Did not converge

iter	Fx	x_1	x_2	Lambda
0	6.4965	0.1	0.1	0.02
1	0.8474	2.1033	0.38002	0.08
2	0.026262	2.5654	0.36012	0.04
3	0.0070726	2.5629	0.4872	0.02
4	2.1415e-05	2.9847	0.4966	0.01
5	9.7715e-08	2.999	0.49971	0.005
6	9.5873e-11	3	0.49999	0.0025

4 . 3 dof
5 . -2 dof
dof 2

n6-A vs n driver
 $n^2 < n^4$

Question 21
Not yet answered
Marked out of 1.00
Flag question



Above is shown a mechanical system consisting of four rigid bodies (numbered 1-4). Two of the bodies are connected to ground through translational joints and the bodies are connected through three revolute joints (named A and C). On the right hand side, a laboratory setup is indicated, where the trajectories of two markers (the blue dots on the mechanical system) are measured in the global coordinate system, with the measurements being denoted $M_1(t)$ and $M_2(t)$. The ultimate goal of the set of questions coming is to setup an optimization problem that can be used to compute the movement of the mechanical system based on the measured marker trajectories. To this end, we wish to use a least-squares approach based on a full Cartesian formulation.

For each body, local coordinate systems are introduced, indicated in blue, red, green and orange. The black coordinate system indicates the global coordinate system. For the i th body, a vector, $\begin{bmatrix} a_i \\ b_i \end{bmatrix}$, indicates the location of the origin of the local coordinate system relative to the global coordinate system. The rotation of the i th body is given by an angle θ_i (not shown in the figure) and used to define the rotation matrix for the body, $A_i = \begin{pmatrix} \cos \theta_i & -\sin \theta_i \\ \sin \theta_i & \cos \theta_i \end{pmatrix}$. The orientation of bodies 1 and 4 (i.e. θ_1 and θ_4) are defined such that the angles are zero in the situation shown in the figure. Additionally, in the figure local vectors from the origins of the local coordinate systems to the joint centers are also defined. For instance, $s_2^{(A)}$ indicates the local coordinates of joint A in body 2's local coordinate system. Similar vectors are indicated for the two markers.

With this information, answer the questions below.

What is the number of unknown coordinates of the problem when using a full Cartesian formulation?

- Select one:
- ☐ A. 2
- ☐ B. 6
- ☐ C. 5
- ☐ D. 4
- ☐ E. 24
- ☐ F. None of the other solutions
- ☒ G. 12

4*3 = 12, lecture 13 slide 10-11 video

3 unknown for each body

$$\phi = \begin{bmatrix} \theta_1 \\ \theta_2 \\ r_1 + A_1 s_2^{(A)} - (r_2 + A_2 s_3^{(A)}) \\ r_4 - (r_3 + A_3 s_3^{(C)}) \\ \theta_4 \end{bmatrix} \begin{matrix} 1 \\ 1 \\ 2 \\ 2 \\ 1 \end{matrix}$$

Question 22
Not yet answered
Marked out of 1.00
Flag question

What is the number of constraint equations required to describe the joints of the mechanical system?

- Select one:
- ☐ A. 10
- ☒ B. 7
- ☐ C. 12
- ☐ D. 11
- ☐ E. 6
- ☐ F. 8
- ☐ G. 5
- ☐ H. 9
- ☐ I. 4
- ☐ J. None of the other solutions

1
3 joints 2 translational
7 equations

ses også her

Question 23
Not yet answered
Marked out of 1.00
Flag question

What is the number of kinematic driver constraint equations that can be formulated from the marker trajectories?

Select one:
☐ A. 5
☒ B. 4
☐ C. 2
☐ D. 3
☐ E. 1
☐ F. 6
☐ G. None of the other solutions

$2 \text{ marker} \cdot x \text{ or } y = 4$

Question 24
Not yet answered
Marked out of 1.00
Flag question

What is the kinematic determinacy of the set of kinematic constraint equations when driving the mechanism with the marker trajectories?

Select one:
☒ A. Kinetically over-determinate
☐ B. Kinetically determinate
☐ C. Kinetically under-determinate

$n^{\text{dof}} \text{ vs } n^{\text{drivers}}$
 $n^{\text{dof}} < n^{\text{drivers}}$

Question 25
Not yet answered
Marked out of 4.00
Flag question

Which of the following choices correspond to the **over-constrained** for the problem when we do not allow violations within the mechanical joint?

Select one:
☐ A. None of the other solutions

☒ B. $\Phi = \begin{pmatrix} p_1 \\ p_2 \\ p_3 \\ p_4 \\ p_5 \end{pmatrix} = \begin{pmatrix} r_1 - (r_2 + A_2 s_2^{(0)}) \\ r_2 + A_2 s_2^{(0)} - (r_3 + A_3 s_3^{(0)}) \\ r_3 - (r_2 + A_2 s_2^{(0)}) \\ r_4 - (r_3 + A_3 s_3^{(0)}) \\ r_5 - (r_2 + A_2 s_2^{(0)}) \end{pmatrix} = 0$

☐ C. $\Phi = \begin{pmatrix} p_1 \\ p_2 \\ p_3 \\ p_4 \end{pmatrix} = \begin{pmatrix} r_1 - (r_2 + A_2 s_2^{(0)}) \\ r_2 + A_2 s_2^{(0)} - (r_3 + A_3 s_3^{(0)}) \\ r_3 - (r_2 + A_2 s_2^{(0)}) \\ r_4 - (r_3 + A_3 s_3^{(0)}) \end{pmatrix} = 0$

☐ D. $\Phi = \begin{pmatrix} p_1 \\ p_2 \\ p_3 \\ p_4 \\ p_5 \end{pmatrix} = \begin{pmatrix} r_1 - (r_2 + A_2 s_2^{(0)}) \\ r_2 + A_2 s_2^{(0)} - (r_3 + A_3 s_3^{(0)}) \\ r_3 - (r_2 + A_2 s_2^{(0)}) \\ r_4 - (r_3 + A_3 s_3^{(0)}) \\ r_5 - (r_2 + A_2 s_2^{(0)}) \end{pmatrix} = 0$

☐ E. $\Phi = \begin{pmatrix} p_1 \\ p_2 \\ p_3 \\ p_4 \end{pmatrix} = \begin{pmatrix} r_1 - (r_2 + A_2 s_2^{(0)}) \\ r_2 + A_2 s_2^{(0)} - (r_3 + A_3 s_3^{(0)}) \\ r_3 - (r_2 + A_2 s_2^{(0)}) \\ r_4 - (r_3 + A_3 s_3^{(0)}) \end{pmatrix} = 0$

☐ F. $\Phi = (B) = 0$

☐ G. $\Phi = \begin{pmatrix} r_2 + A_2 s_2^{(0)} - M_1(t) \\ r_3 + A_3 s_3^{(0)} - M_2(t) \end{pmatrix} = 0$

☐ H. $\Phi = \begin{pmatrix} p_1 \\ p_2 \\ p_3 \\ p_4 \end{pmatrix} = 0$

$\Phi = \begin{pmatrix} y_1 \\ \theta_1 \\ r_1 - (r_2 + A_2 s_2^{(A)}) \\ r_2 + A_2 s_2^{(0)} - (r_3 + A_3 s_3^{(A)}) \\ r_4 - (r_3 + A_3 s_3^{(A)}) \\ x_4 \\ \theta_4 \end{pmatrix}$

no!

Question 26
Not yet answered
Marked out of 4.00
Flag question

Which of the following choices correspond to the **over-constrained** for the problem when we do not allow violations within the mechanical joint?

Select one:
☐ A. $\Phi = \begin{pmatrix} r_2 + A_2 s_2^{(0)} \\ r_3 + A_3 s_3^{(0)} \end{pmatrix}$

☐ B. $\Phi = \begin{pmatrix} r_2 + A_2 s_2^{(0)} - M_1(t) \\ r_3 + A_3 s_3^{(0)} - M_2(t) \\ \theta_1 - \omega_1 t \\ \theta_2 - \omega_2 t \\ \theta_3 - \omega_3 t \\ \theta_4 - \omega_4 t \end{pmatrix}$

☐ C. $\Phi = \begin{pmatrix} r_1 \\ \theta_1 \\ r_2 \\ \theta_2 \end{pmatrix}$

☒ D. $\Phi = \begin{pmatrix} r_2 + A_2 s_2^{(0)} - M_1(t) \\ r_3 + A_3 s_3^{(0)} - M_2(t) \end{pmatrix}$

☐ E. $\Phi = \begin{pmatrix} p_1 \\ p_2 \\ p_3 \\ p_4 \\ p_5 \end{pmatrix} = \begin{pmatrix} r_1 - (r_2 + A_2 s_2^{(0)}) \\ r_2 + A_2 s_2^{(0)} - (r_3 + A_3 s_3^{(0)}) \\ r_3 - (r_2 + A_2 s_2^{(0)}) \\ r_4 - (r_3 + A_3 s_3^{(0)}) \\ r_5 - (r_2 + A_2 s_2^{(0)}) \end{pmatrix} = 0$

☐ F. $\Phi = \begin{pmatrix} \theta_1 - \omega_1 t \\ \theta_2 - \omega_2 t \\ \theta_3 - \omega_3 t \\ \theta_4 - \omega_4 t \end{pmatrix}$

☐ G. None of the other solutions

☐ H. $\Phi = \begin{pmatrix} \theta_1 - \omega_1 t \\ \theta_2 - \omega_2 t \\ \theta_3 - \omega_3 t \\ \theta_4 - \omega_4 t \end{pmatrix}$

$\Psi = \begin{pmatrix} r_2 + A_2 s_2^{(A)} - M_1(t) \\ r_3 + A_3 s_3^{(A)} - M_2(t) \end{pmatrix}$

Question 27
Not yet answered
Marked out of 3.00
Flag question

Which of the following optimization problem formulations enable the computation of the movement of the mechanism with the specified hard and soft constraints?

Select one:
☐ A. None of the other solutions

☒ B. $\min \frac{1}{2} \Psi^T \Psi$
s.t. $\Phi = 0$

☐ C. $\min \frac{1}{2} \Phi^T \Phi$

☐ D. $\Phi = \begin{pmatrix} r_2 + A_2 s_2^{(0)} - M_1(t) \\ r_3 + A_3 s_3^{(0)} - M_2(t) \\ \theta_1 - \omega_1 t \\ \theta_2 - \omega_2 t \\ \theta_3 - \omega_3 t \\ \theta_4 - \omega_4 t \end{pmatrix}$

☐ E. $\min \frac{1}{2} \Phi^T \Phi$
s.t. $\Phi = 0$

☐ F. $\min \frac{1}{2} \Psi^T \Psi$

$\min \frac{1}{2} \Psi^T \Psi$
s.t. $\Phi = 0$

1:12 = 12.22